Chapter 3. Probability and Probability Distributions

Probability, random variables, expectation, covariance

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3.1 Definition of probability

Motivation. A survey observes part of a population and draws conclusions about the whole by induction; probability is the logical basis that makes such conclusions quantifiable. We first fix the objects to which probabilities are assigned: sample spaces and events.

Definition (Sample space and events)

The set of all possible outcomes of a statistical survey (experiment) is the **sample space** S. A subset of S—the set of outcomes satisfying some condition of interest—is an **event**; the event A *occurs* if the observed outcome is an element of A. An event consisting of a single element is an **elementary** (**simple**) **event**.
$A \cup B$, $A \cap B$, A^c are the **union**, **intersection** and **complement**. If $A \cap B = \varnothing$, A and B are **mutually disjoint** (exclusive): they cannot occur together.

Illustration 3-1: sample spaces

(a) Number shown by one die: $S = \{1, 2, 3, 4, 5, 6\}$.
(b) Number of items produced until the 10th defective: $S = \{10, 11, 12, \dots\}$.
(c) Lifetime of a light bulb drawn from a factory: $S = \{t \mid t \geq 0\}$.

When each of finitely many elementary events is equally likely (as for the six faces of a die), Laplace defined probability as follows.

Definition 3-1 (Classical definition of probability)

If $S = \{e_1, \dots, e_N\}$ has N elements whose elementary events are equally likely, the **probability** of an event A with m elements is

$$P(A) = \frac{m}{N}.$$

Example 3-1 (Table-tennis balls)

A box contains 3 white balls and 1 yellow ball. Two balls are drawn by simple random sampling. Find the probability of getting one white and one yellow ball.
Solution. Label the white balls a, b, c and the yellow one d:

$$S = \{\{a,b\},\{a,c\},\{a,d\},\{b,c\},\{b,d\},\{c,d\}\}, \qquad A = \{\{a,d\},\{b,d\},\{c,d\}\}.$$

All 6 pairs are equally likely, so $P(A) = 3/6 = 1/2$.

The classical definition fails when outcomes are infinitely many or not equally likely. Experience shows that, as a survey is repeated, the *relative frequency* of an event tends to settle at a constant (Figure 3-1: the proportion of defectives approaches a constant as production grows), which can be called the probability of the event. Based on the basic properties of relative frequencies, Kolmogorov gave the axioms.

Definition 3-2 (Axiomatic definition of probability)

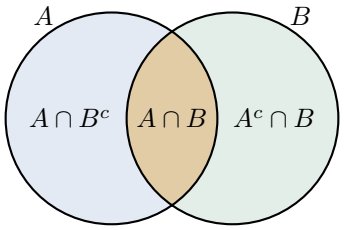
A rule P assigning numbers to events of S is a probability if
(1) $0 \leq P(A) \leq 1$ for every event A;
(2) $P(S) = 1$;

(3) for mutually disjoint events $A_1, A_2, \ldots,$

$$P(A_1 \cup A_2 \cup \cdots) = P(A_1) + P(A_2) + \cdots.$$

$P(A)$ is called the **probability** of A.

This covers empirical probability (limits of relative frequencies) as well as subjective probability (values set by personal judgement).



Partition of $A \cup B$ into three disjoint pieces (Figure 3-2)

Theorem 3-1 (Properties of probability)

(1) $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
(2) $P(A^c) = 1 - P(A)$.
(3) If $A \subset B$, then $P(A) \leq P(B)$.

Proof. (1) $A \cap B^c$, $A \cap B$, $A^c \cap B$ are mutually disjoint, with $A \cup B$ their union, $A = (A \cap B^c) \cup (A \cap B)$ and $B = (A \cap B) \cup (A^c \cap B)$. By axiom (3),

$$\begin{aligned} P(A \cup B) &= P(A \cap B^c) + P(A \cap B) + P(A^c \cap B), \\ P(A) &= P(A \cap B^c) + P(A \cap B), \\ P(B) &= P(A \cap B) + P(A^c \cap B). \end{aligned}$$

Subtracting, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

(2) A and A^c are disjoint with union S, so $P(A) + P(A^c) = P(S) = 1$ by axioms (3) and (2).

(3) If $A \subset B$, then $B = A \cup (A^c \cap B)$ is a disjoint union, so $P(B) = P(A) + P(A^c \cap B) \geq P(A)$ by axioms (3) and (1). ■

Example 3-2 (Three bags)

Each of three bags contains 5 marbles numbered 1 to 5. One marble is drawn from each bag. Find the probability that the sum of the numbers is at least 5.
Solution. Let $A =$ "sum ≥ 5"; then $A^c =$ "sum ≤ 4" $= B \cup C$ with $B =$ "sum $= 3$", $C =$ "sum $= 4$", $B \cap C = \varnothing$. There are $5^3 = 125$ equally likely outcomes, and

$$B = \{(1,1,1)\}, \qquad C = \{(1,1,2), (1,2,1), (2,1,1)\},$$

so $P(B) = 1/125$, $P(C) = 3/125$ and

$$P(A) = 1 - \{P(B) + P(C)\} = 1 - \frac{4}{125} = \frac{121}{125}.$$

3.2 Conditional probability and independence

Motivation. Information that one event has occurred changes how likely another is. Conditional probability formalizes this update, and independence describes when no update is needed.

Given that A has occurred, outcomes in A^c are impossible, so A plays the role of the whole sample space; the probability that B occurs is the share of $P(A)$ taken by $P(A \cap B)$.

Definition 3-3 (Conditional probability)

The **conditional probability** of B given A is

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)} \qquad (P(A) > 0).$$

Example 3-3 (Two dice)

Two dice are thrown. Given that the first die shows more than the second, find the probability that the sum is 7.
Solution. Let $A =$ "first $>$ second", $B =$ "sum $= 7$". Of 36 outcomes, A has 15, so $P(A) = 15/36$. Also $B = \{(1,6),(2,5),(3,4),(4,3),(5,2),(6,1)\}$ and $A \cap B = \{(4,3),(5,2),(6,1)\}$, so

$$P(B) = \frac{6}{36} = \frac{1}{6}, \qquad P(A \cap B) = \frac{3}{36} = \frac{1}{12}, \qquad P(B \mid A) = \frac{P(A \cap B)}{P(A)} = \frac{3}{15} = \frac{1}{5}.$$

Theorem 3-2 (Multiplication rule)

If $P(A) > 0$ and $P(B) > 0$, then

$$P(A \cap B) = P(A \mid B)P(B) = P(B \mid A)P(A).$$

Proof. By Definition 3-3, $P(B \mid A) = P(A \cap B)/P(A)$; multiply by $P(A)$. Exchanging the roles of A and B (and using $B \cap A = A \cap B$) gives $P(A \cap B) = P(A \mid B)P(B)$. ■

The rule is useful when conditional probabilities are easy to compute.

Example 3-4 (Prize draw)

A couple joins a draw at a supermarket: a box holds 10 winning and 50 losing balls; each person draws once and a drawn ball is not returned. Find the probability that both win.
Solution. Let $A =$ "husband wins", $B =$ "wife wins". Then

$$P(A \cap B) = P(B \mid A)P(A) = \frac{9}{59} \times \frac{10}{60} = \frac{3}{118}.$$

When $P(B)$ is hard to compute directly, split S into mutually disjoint events $A_1, \ldots, A_n$ with $A_1 \cup \cdots \cup A_n = S$ (a *partition*). Then the $B \cap A_i$ are mutually disjoint with union B.

Theorem 3-3 (Law of total probability)

If $A_i \cap A_j = \varnothing$ $(i \neq j)$, $A_1 \cup A_2 \cup \cdots \cup A_n = S$ and $P(A_i) > 0$, then

$$P(B) = P(B \mid A_1)P(A_1) + \cdots + P(B \mid A_n)P(A_n).$$

Proof.

Step 1. For $i \neq j$, $(B \cap A_i) \cap (B \cap A_j) \subset A_i \cap A_j = \varnothing$, and

$$(B \cap A_1) \cup \cdots \cup (B \cap A_n) = B \cap (A_1 \cup \cdots \cup A_n) = B \cap S = B.$$

Step 2. By axiom (3), $P(B) = \sum_{i=1}^{n} P(B \cap A_i)$, and by Theorem 3-2 each term equals $P(B \mid A_i)P(A_i)$. ■

Example 3-5 (Joint training)

Three football teams train together: 50% of the players are from team 1, 30% from team 2, 20% from team 3; 20%, 15%, 30% of the respective teams are forwards. A player is chosen at random. Find the probability that he is a forward.
Solution. With A_i = "from team i" (a partition) and B = "forward",

$$P(B) = 0.2 \times 0.5 + 0.15 \times 0.3 + 0.3 \times 0.2 = 0.205.$$

If the occurrence of A does not affect the probability of B, i.e. $P(B \mid A) = P(B)$, then A and B are **(mutually) independent**. Then $P(B) = P(A \cap B)/P(A)$, i.e. $P(A \cap B) = P(A)P(B)$; and by the multiplication rule $P(A)P(B) = P(A \mid B)P(B)$, i.e. $P(A \mid B) = P(A)$.

Definition 3-4 (Independent events)

Events A and B are **independent** if one of the following holds (when the conditional probabilities are defined, they are equivalent):

$$P(A \cap B) = P(A)P(B), \qquad P(A \mid B) = P(A), \qquad P(B \mid A) = P(B).$$

Example 3-6 (Prize draw revisited)

In Example 3-4:
(a) Are "husband wins" and "wife wins" independent?
(b) Are they independent if the drawn ball is returned to the box?
Solution.
(a) By the law of total probability,

$$P(B) = P(B \mid A)P(A) + P(B \mid A^c)P(A^c) = \frac{9}{59} \cdot \frac{10}{60} + \frac{10}{59} \cdot \frac{50}{60} = \frac{1}{6}.$$

Since $P(B) \neq P(B \mid A) = 9/59$, A and B are not independent.
(b) With replacement, $P(B \mid A) = 10/60 = 1/6 = P(B)$, so A and B are independent.

Under simple random sampling with replacement successive draws are independent; without replacement they are not. Events that are not independent are **(mutually) dependent**. In general, $A_1, \ldots, A_n$ are **independent** if

$$\begin{aligned} P(A_i \cap A_j) &= P(A_i)P(A_j) && (1 \le i < j \le n), \\ P(A_i \cap A_j \cap A_k) &= P(A_i)P(A_j)P(A_k) && (1 \le i < j < k \le n), \\ &\vdots \\ P(A_1 \cap A_2 \cap \cdots \cap A_n) &= P(A_1)P(A_2)\cdots P(A_n). \end{aligned}$$

3.3 Random variables and probability distributions

Motivation. Outcomes may be qualitative (blood types O, AB, A, B), which makes probability calculations clumsy. Mapping outcomes to numbers turns probability on the sample space into probability on sets of numbers.

Illustration 3-2: coin toss

Tossing one coin, $S = \{\mathrm{H}, \mathrm{T}\}$ with $P(\mathrm{H}) = P(\mathrm{T}) = 1/2$. Define $X(\mathrm{H}) = 1$, $X(\mathrm{T}) = 0$. Then X is a function on S, and the events $\{X = 0\}$, $\{X = 1\}$ each have probability $1/2$: X transfers the probability on S to the set $\{0, 1\}$.

Definition (Random variable, distribution)

A real-valued function defined on a sample space is a **random variable**. If an outcome is mapped to the value a, we say "X takes the value a". The description of how probability is spread over the values of X, by nonnegative numbers summing to 1, is the **probability distribution** (or simply **distribution**) of X.

Illustration 3-3

In Illustration 3-2 the distribution of X is $P(X = 0) = P(X = 1) = 1/2$.

If the possible values $x_1, x_2, \dots$ of X can be listed (countable), X is a **discrete random variable**, and

$$p(x) = \begin{cases} P(X = x_i), & x = x_i \ (i = 1, 2, \dots) \\ 0, & \text{otherwise} \end{cases}$$

is its **probability mass function**.

Definition 3-5 (Probability mass function of a discrete r.v.)

A function $p(x)$ is the **probability mass function** of the discrete random variable X if
(1) $0 \le p(x) \le 1$ and $\sum_{\text{all } x} p(x) = 1$;
(2) $P(a < X \le b) = \sum_{a < x \le b} p(x)$.

Thus the distribution of a discrete random variable is completely determined by its probability mass function; with few values it is convenient to list it as a table.

Example 3-7 (Popular posts)

Of an influencer's 12 posts, 4 have more than 10,000 likes. Three posts are chosen by simple random sampling; X is the number among them with more than 10,000 likes. Find the distribution of X.
Solution. There are $\binom{12}{3} = 220$ equally likely choices, and $p(x) = \binom{8}{3-x}\binom{4}{x}/\binom{12}{3}$:

$$p(0) = \frac{56 \times 1}{220} = \frac{14}{55}, \qquad p(1) = \frac{28 \times 4}{220} = \frac{28}{55},$$
$$p(2) = \frac{8 \times 6}{220} = \frac{12}{55}, \qquad p(3) = \frac{1 \times 4}{220} = \frac{1}{55}.$$

If X can take every value in an interval (so its values cannot be listed), X is a **continuous random variable**.

Illustration 3-4: spinner

A pointer fixed at the centre of a dial marked from 0 to 1 spins freely; X is the mark where it stops. X can take any value in $[0, 1]$.

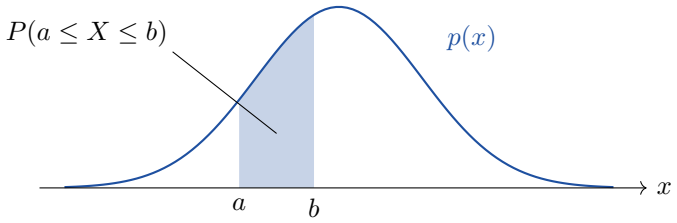
The probability that the pointer stops in $[a, b]$ is the ratio of the lengths:

$$P(a \le X \le b) = \frac{b-a}{1-0} = b - a \qquad (0 \le a < b \le 1).$$

With $p(x) = 1$ for $0 \le x \le 1$ and $p(x) = 0$ otherwise, this is $P(a \le X \le b) = \int_a^b p(x)\,dx$. So probabilities for X are determined by p, its **probability density function**.

Definition 3-6 (Probability density function of a continuous r.v.)

A function $p(x)$ is the **probability density function** of the continuous random variable X if
(1) $p(x) \ge 0$ and $\int_{-\infty}^{\infty} p(x)\,dx = 1$;
(2) $P(a \le X \le b) = \int_a^b p(x)\,dx$ $(-\infty \le a < b \le \infty)$.



$P(a \le X \le b)$ is the area under the density over $[a, b]$ (Figure 3-5)

Theorem 3-4 (Properties of continuous random variables)

For a continuous random variable X,
(1) $P(X = c) = 0$ for every c;
(2) $P(a \le X \le b) = P(a \le X < b) = P(a < X \le b) = P(a < X < b)$.

Proof. (1) $P(X = c)$ is the area at the single point c. Precisely, for every $h > 0$, $\{X = c\} \subset \{c - h \le X \le c + h\}$, so by Theorem 3-1(3)

$$0 \le P(X = c) \le \int_{c-h}^{c+h} p(x)\,dx \longrightarrow 0 \qquad (h \to 0),$$

since the integral over a shrinking interval of an integrable function tends to 0.

(2) $\{a \le X \le b\}$ is the disjoint union of $\{a < X < b\}$ with (some of) $\{X = a\}$ and $\{X = b\}$, which have probability 0 by (1); apply axiom (3). ■

Example 3-8 (Finding the constant)

A continuous random variable X has density $p(x) = bx(1 - x)$ for $0 \le x \le 1$, and 0 for $x < 0$ or $x > 1$. Find b and $P(0 \le X \le \frac{2}{3})$.
Solution. $1 = \int_0^1 bx(1-x)\,dx = b/6$, so $b = 6$. Then

$$P\Big(0 \le X \le \frac{2}{3}\Big) = \int_0^{2/3} 6x(1-x)\,dx = \Big[3x^2 - 2x^3\Big]_0^{2/3} = \frac{4}{3} - \frac{16}{27} = \frac{20}{27}.$$

When one characteristic of a population is observed, the sample space is the set of distinct characteristic values; the observed value is a random variable whose distribution equals the population distribution. So, to model a population theoretically, we use a random variable with the same distribution.

3.4 Expected value and its properties

Motivation. Chapter 2 summarized populations by mean and standard deviation. The same summaries for a probability distribution are the expected value and the variance of a random variable.

If X takes $x_1, x_2, \ldots$ with p.m.f. p, the **mean** or **expected value** $E(X)$ is $\sum_i x_i p(x_i) = x_1 P(X = x_1) + x_2 P(X = x_2) + \cdots$; for a density p it is $\int x p(x)\,dx$.

Definition 3-7 (Expected value of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$,

$$E(X) = \begin{cases} \sum_{\text{all } x} x\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Example 3-9 (A mean)

X has distribution $p(1) = \frac{1}{8}$, $p(2) = \frac{3}{8}$, $p(3) = \frac{3}{8}$, $p(4) = \frac{1}{8}$. Find $E(X)$.
Solution. $E(X) = 1 \cdot \frac{1}{8} + 2 \cdot \frac{3}{8} + 3 \cdot \frac{3}{8} + 4 \cdot \frac{1}{8} = 2.5$, the centre of the distribution graph (Figure 3-6).

As in mechanics, if masses $p(x_i)$ sit at $x_1, x_2, \ldots$, then $\sum x_i p(x_i)$ is the centre of mass relative to the origin; likewise $\int x p(x)\,dx$ for a mass density p. If the distribution of X models a population, $E(X)$ equals the population mean, often written μ.

Example 3-10 (A function of X)

A coin is tossed 3 times; X is the number of heads.
(a) Find the distribution of $Y = X(3 - X)$.
(b) Find $E(Y)$.
Solution.
(a) $p_X(0) = \frac{1}{8}$, $p_X(1) = \frac{3}{8}$, $p_X(2) = \frac{3}{8}$, $p_X(3) = \frac{1}{8}$. $Y = 0$ when $X \in \{0, 3\}$ and $Y = 2$ when $X \in \{1, 2\}$, so $p_Y(0) = \frac{1}{4}$, $p_Y(2) = \frac{3}{4}$.
(b) $E[X(3 - X)] = E(Y) = 0 \cdot \frac{1}{4} + 2 \cdot \frac{3}{4} = \frac{3}{2}$.

A function of a random variable is again a random variable. Its expectation can be computed without first finding its distribution:

$$\begin{aligned} E[X(3 - X)] &= 0(3 - 0)p_X(0) + 1(3 - 1)p_X(1) + 2(3 - 2)p_X(2) + 3(3 - 3)p_X(3) \\ &= 0 \cdot \tfrac{1}{8} + 2 \cdot \tfrac{3}{8} + 2 \cdot \tfrac{3}{8} + 0 \cdot \tfrac{1}{8} = \tfrac{3}{2}. \end{aligned}$$

Theorem 3-5 (Expectation of a function of a random variable)

If the p.m.f. (p.d.f.) of the discrete (continuous) random variable X is $p(x)$, then for any function g

$$E[g(X)] = \begin{cases} \sum_{\text{all } x} g(x)\,p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} g(x)\,p(x)\,dx & \text{(continuous)} \end{cases}$$

Proof. [Supplement]

Step 1 (discrete case). $Y = g(X)$ is discrete; for each value y, $\{Y = y\}$ is the disjoint union of $\{X = x\}$ over the x with $g(x) = y$, so $p_Y(y) = \sum_{x:\,g(x)=y} p(x)$. Hence

$$E(Y) = \sum_y y\,p_Y(y) = \sum_y \sum_{x:\,g(x)=y} g(x)\,p(x) = \sum_{\text{all } x} g(x)\,p(x),$$

since every x appears in exactly one group (the rearrangement is valid when the sum converges absolutely, which is the condition for $E(Y)$ to exist).

Step 2 (continuous case, monotone g). Let g be differentiable and strictly increasing with inverse h. Then $P(Y \le y) = P(X \le h(y)) = \int_{-\infty}^{h(y)} p(x)\,dx$, and differentiating, Y has density $p_Y(y) = p(h(y))h'(y)$. Substituting $x = h(y)$,

$$E(Y) = \int y\,p(h(y))h'(y)\,dy = \int g(x)\,p(x)\,dx.$$

For strictly decreasing g, $p_Y(y) = -p(h(y))h'(y)$ and the substitution reverses the limits, giving the same result.

Step 3 (general g). For general g split the line into intervals on which g is monotone and apply Step 2 on each piece; for completely arbitrary g the formula is the general change-of-variables rule of measure-theoretic integration, which we accept. ■

Example 3-11 (Waiting for a bus)

Buses leave a stop at 0, 15, 30 and 45 minutes past each hour. A person arrives at a random time. Find the expected waiting time.

Solution. Let X (minutes past the hour) be the arrival time; X has constant density $p(x) = \frac{1}{60}$ on $0 < x \le 60$. The waiting time is

$$g(x) = 15 - x, \quad 30 - x, \quad 45 - x, \quad 60 - x \qquad \text{on } (0, 15],\ (15, 30],\ (30, 45],\ (45, 60].$$

By Theorem 3-5,

$$\begin{aligned} E[g(X)] &= \frac{1}{60}\Big\{ \int_0^{15} (15 - x)\,dx + \int_{15}^{30} (30 - x)\,dx + \int_{30}^{45} (45 - x)\,dx + \int_{45}^{60} (60 - x)\,dx \Big\} \\ &= \frac{1}{60} \times 4 \times 112.5 = 7.5, \end{aligned}$$

i.e. 7 minutes 30 seconds.

Theorem 3-6 (Properties of expected value)

(1) $E(aX + b) = aE(X) + b$ (a, b constants).
(2) $E[c_1 g_1(X) + c_2 g_2(X)] = c_1 E[g_1(X)] + c_2 E[g_2(X)]$ (c_1, c_2 constants, g_1, g_2 any functions).

Proof. [Supplement] (2) By Theorem 3-5, in the discrete case

$$E[c_1 g_1(X) + c_2 g_2(X)] = \sum_x \{c_1 g_1(x) + c_2 g_2(x)\} p(x) = c_1 \sum_x g_1(x) p(x) + c_2 \sum_x g_2(x) p(x),$$

which is $c_1 E[g_1(X)] + c_2 E[g_2(X)]$; the continuous case is identical with integrals, using linearity of the integral.

(1) Take $g_1(x) = x$, $g_2(x) = 1$, $c_1 = a$, $c_2 = b$ in (2), and note $E(1) = \sum_x p(x) = 1$ (or $\int p = 1$). ■

The **standard deviation** of X (mean μ) is $\mathrm{sd}(X) = \sqrt{E[(X-\mu)^2]}$, and its square $E[(X-\mu)^2]$ is the **variance** $\mathrm{Var}(X)$.

Definition 3-8 (Variance and standard deviation of a random variable)

If X has mean μ and p.m.f. (or p.d.f.) $p(x)$,

(1) $\mathrm{Var}(X) = E[(X-\mu)^2] = \begin{cases} \sum_{\text{all } x} (x-\mu)^2 p(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} (x-\mu)^2 p(x)\, dx & \text{(continuous)} \end{cases}$

(2) $\mathrm{sd}(X) = \sqrt{\mathrm{Var}(X)} = \sqrt{E[(X-\mu)^2]}$.

When the distribution of X models a population, $\mathrm{Var}(X)$ and $\mathrm{sd}(X)$ are the population variance and standard deviation, often written σ^2 and σ.

Theorem 3-7 (Shortcut formula for the variance)

$$\mathrm{Var}(X) = E(X^2) - [E(X)]^2.$$

Proof. By Theorem 3-6(2) with $\mu = E(X)$,

$$\mathrm{Var}(X) = E[(X-\mu)^2] = E(X^2 - 2\mu X + \mu^2) = E(X^2) - 2\mu E(X) + \mu^2 = E(X^2) - \mu^2.$$

■

Example 3-12 (Two coin tosses)

X is the number of heads in 2 tosses of a coin. Find the mean and standard deviation of X.
Solution.

x	0	1	2	Total
$p(x)$	1/4	1/2	1/4	1
$xp(x)$	0	1/2	1/2	1
$x^2 p(x)$	0	1/2	1	3/2

So $E(X) = 1$, $E(X^2) = \frac{3}{2}$, $\mathrm{Var}(X) = \frac{3}{2} - 1^2 = \frac{1}{2}$ and $\mathrm{sd}(X) = 1/\sqrt{2}$.

Example 3-13 (Spinner)

For the spinner of Illustration 3-4, find the mean and standard deviation of X.
Solution. $p(x) = 1$ on $[0,1]$, so

$$E(X) = \int_0^1 x\, dx = \frac{1}{2}, \qquad E(X^2) = \int_0^1 x^2\, dx = \frac{1}{3},$$

$$\mathrm{Var}(X) = \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12}, \qquad \mathrm{sd}(X) = \frac{1}{\sqrt{12}}.$$

Theorem 3-8 (Property of the variance)

$$\mathrm{Var}(aX + b) = a^2 \,\mathrm{Var}(X) \qquad (a, b \text{ constants}).$$

Proof. [Supplement] Let $\mu = E(X)$. By Theorem 3-6(1), $E(aX+b) = a\mu + b$, so $(aX+b) - E(aX+b) = a(X-\mu)$ and

$$\mathrm{Var}(aX + b) = E[a^2(X-\mu)^2] = a^2 E[(X-\mu)^2] = a^2 \,\mathrm{Var}(X).$$

■

Subtracting the mean and dividing by the standard deviation is **standardization**:

$$Z = \frac{X - E(X)}{\mathrm{sd}(X)}.$$

By Theorems 3-6 and 3-8 (with $a = 1/\mathrm{sd}(X)$, $b = -E(X)/\mathrm{sd}(X)$), $E(Z) = 0$ and $\mathrm{Var}(Z) = \mathrm{Var}(X)/\mathrm{sd}(X)^2 = 1$.

3.5 Joint distribution of two random variables

Motivation. Height and weight of a newborn, or English and maths scores, are observed together and are related. The individual distributions do not capture that relation; the joint distribution does.

We treat discrete random variables; the results extend to the continuous case.

Illustration 3-5: same marginals, different relations

Three different coins A, B, C are tossed. Let $X = 1$ if A shows heads (else 0); $Y = 1$ if A and B both show heads (else 0); $Z = 1$ if B and C both show heads (else 0). Y and Z have the same distribution, $P(0) = \frac{3}{4}$, $P(1) = \frac{1}{4}$. Yet the relation of X to Y differs from that of X to Z: $X = 0$ forces $Y = 0$, whereas given $X = 0$, Z is 0 or 1 with probabilities $\frac{3}{4}, \frac{1}{4}$.

Definition 3-9 (Joint distribution of two random variables)

The description of how probability is spread over all ordered pairs of values of X and Y, by nonnegative numbers summing to 1, is the **joint probability distribution** (or **joint distribution**) of X and Y.

For discrete X, Y with values $x_1, x_2, \ldots$ and $y_1, y_2, \ldots$,

$$p(x,y) = \begin{cases} P(X = x,\, Y = y), & (x,y) = (x_i, y_j)\ (i,j = 1,2,\ldots) \\ 0, & \text{otherwise} \end{cases}$$

is the **joint probability mass function**.

Definition 3-10 (Joint probability mass function)

$p(x,y)$ is the **joint probability mass function** of the discrete random variables X and Y if
(1) $0 \le p(x,y) \le 1$ and $\sum_{\text{all } x}\sum_{\text{all } y} p(x,y) = 1$;
(2) $P(a < X \le b,\ c < Y \le d) = \sum_{a<x\le b}\sum_{c<y\le d} p(x,y)$.

Example 3-14 (Joint distributions from three coins)

For the experiment of Illustration 3-5, find the joint distribution of
(a) X and Y; (b) X and Z.
Solution. Listing coins A, B, C:

Outcome	HHH	HHT	HTH	HTT	THH	THT	TTH	TTT
X	1	1	1	1	0	0	0	0
Y	1	1	0	0	0	0	0	0
Z	1	0	0	0	1	0	0	0

Each outcome has probability $\frac{1}{8}$; collecting them,

(a) X and Y

$y\backslash x$	0	1	Row sum
0	2/4	1/4	3/4
1	0	1/4	1/4
Col. sum	2/4	2/4	1

(b) X and Z

$z\backslash x$	0	1	Row sum
0	3/8	3/8	6/8
1	1/8	1/8	2/8
Col. sum	4/8	4/8	1

The column and row sums reproduce the distributions of X and Y. In general the **marginal probability mass functions** are

$$p_1(x) = \sum_{\text{all } y} p(x,y), \qquad p_2(y) = \sum_{\text{all } x} p(x,y),$$

and they determine the distributions of X and Y (because $\{X = x\}$ is the disjoint union of the events $\{X = x, Y = y\}$ over y).

Example 3-15 (Two games on one die)

A game pays according to one die roll. A receives 100 won for 1 or 2, 200 won for 3 or 4, 300 won for 5 or 6; B receives 100 won for an even number and (face $\times 100$) won for an odd number. Find the distribution of Z, the total received by A and B.
Solution. Let X, Y be the amounts of A and B. E.g. $(X,Y) = (100,100)$ occurs for faces 1 and 2, so $p(100,100) = \frac{2}{6}$. The joint distribution is

$x\backslash y$	100	300	500	Row sum
100	2/6	0	0	2/6
200	1/6	1/6	0	2/6
300	1/6	0	1/6	2/6
Col. sum	4/6	1/6	1/6	1

$Z = X + Y$ takes the values 200, 300, 400, 500, 800 with probabilities $\frac{2}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}, \frac{1}{6}$.

Theorem 3-9 (Expectation of a function of two random variables)

Let X, Y be discrete with joint p.m.f. $p(x, y)$. For any functions g, g_1, g_2:
(1) $E[g(X,Y)] = \sum_{\text{all } x} \sum_{\text{all } y} g(x,y)\, p(x,y)$;
(2) $E[c_1 g_1(X,Y) + c_2 g_2(X,Y)] = c_1 E[g_1(X,Y)] + c_2 E[g_2(X,Y)]$ (c_1, c_2 constants).

Proof. [Supplement] (1) $W = g(X,Y)$ is discrete, and $\{W = w\}$ is the disjoint union of the events $\{X = x, Y = y\}$ over pairs with $g(x,y) = w$, so $p_W(w) = \sum_{(x,y):\, g(x,y)=w} p(x,y)$. Hence

$$E(W) = \sum_w w\, p_W(w) = \sum_w \sum_{(x,y):\, g(x,y)=w} g(x,y)\, p(x,y) = \sum_x \sum_y g(x,y)\, p(x,y),$$

each pair (x, y) appearing exactly once (absolute convergence justifies the regrouping).

(2) By (1) applied to $c_1 g_1 + c_2 g_2$, then splitting the double sum:

$$\sum_x \sum_y \{c_1 g_1 + c_2 g_2\}(x,y)\, p(x,y) = c_1 \sum_x \sum_y g_1(x,y) p(x,y) + c_2 \sum_x \sum_y g_2(x,y) p(x,y).$$

■

Extending independence of events (Section 3.2): X and Y are independent if, for all x, y, the events $\{X = x\}$ and $\{Y = y\}$ are independent.

Definition 3-11 (Independence of two random variables)

Let $p(x,y)$ be the joint p.m.f. (joint p.d.f.) of discrete (continuous) X, Y and p_1, p_2 the marginals. X and Y are **independent** if for all x, y

$$p(x,y) = p_1(x)\, p_2(y).$$

Otherwise they are **dependent**.

Example 3-16 (Independence check)

A coin is tossed three times; X is the number of heads in the first two tosses and Y the number of heads on the third toss. Are X and Y independent?
Solution. The joint distribution is

$x \backslash y$	0	1	Row sum
0	1/8	1/8	1/4
1	2/8	2/8	1/2
2	1/8	1/8	1/4
Col. sum	1/2	1/2	1

Every cell equals the product of its margins, e.g. $p_1(0)p_2(0) = \frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8} = p(0,0)$ and $p_1(1)p_2(1) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4} = p(1,1)$; the other four cells check the same way. Hence X and Y are independent.

3.6 Covariance and correlation

Motivation. When X and Y tend to be large together, the product of their standardized deviations is typically positive. Averaging this product gives a single number measuring the direction and strength of linear association.

Let X, Y have means μ_1, μ_2 and standard deviations σ_1, σ_2. The **correlation coefficient** $\mathrm{Corr}(X,Y)$ is the expected value of the product of standardized variables $(\frac{X-\mu_1}{\sigma_1})(\frac{Y-\mu_2}{\sigma_2})$. It measures the strength and direction of a *linear* relation, and always $-1 \le \mathrm{Corr}(X,Y) \le 1$. In the extreme case $Y = aX + b$ ($a \neq 0$), the correlation is 1 if $a > 0$ and -1 if $a < 0$. The expected value of $(X-\mu_1)(Y-\mu_2)$, i.e. the correlation times $\sigma_1\sigma_2$, is the **covariance** $\mathrm{Cov}(X,Y)$.

Definition 3-12 (Covariance and correlation coefficient)

(1) $\mathrm{Cov}(X,Y) = E[(X-\mu_1)(Y-\mu_2)]$.

(2) $\mathrm{Corr}(X,Y) = E\Big[\Big(\frac{X-\mu_1}{\sigma_1}\Big)\Big(\frac{Y-\mu_2}{\sigma_2}\Big)\Big] = \dfrac{\mathrm{Cov}(X,Y)}{\mathrm{sd}(X)\,\mathrm{sd}(Y)}$.

Proposition (Range of the correlation)

$-1 \le \mathrm{Corr}(X,Y) \le 1$; and if $Y = aX + b$ with $a \neq 0$, then $\mathrm{Corr}(X,Y) = 1$ for $a > 0$ and -1 for $a < 0$.

Proof. [Supplement]

Step 1. Let $U = (X-\mu_1)/\sigma_1$, $V = (Y-\mu_2)/\sigma_2$ and $\rho = \mathrm{Corr}(X,Y) = E(UV)$. Then $E(U^2) = E(V^2) = 1$, and by Theorem 3-9(2)

$$0 \le E[(U-\rho V)^2] = E(U^2) - 2\rho E(UV) + \rho^2 E(V^2) = 1 - \rho^2,$$

so $\rho^2 \le 1$.

Step 2. If $Y = aX + b$, then $\mu_2 = a\mu_1 + b$ and $\sigma_2 = |a|\sigma_1$ (Theorem 3-8), so $V = \frac{a}{|a|}U$ and $\rho = \frac{a}{|a|}E(U^2) = \pm 1$ according to the sign of a. ■

Theorem 3-10 (Shortcut formula for the covariance)

$$\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y).$$

Proof. By Theorem 3-9(2),

$$\begin{aligned}\mathrm{Cov}(X,Y) &= E[(X-\mu_1)(Y-\mu_2)] = E(XY - \mu_2 X - \mu_1 Y + \mu_1\mu_2)\\ &= E(XY) - \mu_2 E(X) - \mu_1 E(Y) + \mu_1\mu_2 = E(XY) - \mu_1\mu_2.\end{aligned}$$

■

Example 3-17 (Covariance and correlation)

X and Y have the joint distribution below. Find $\mathrm{Cov}(X,Y)$ and $\mathrm{Corr}(X,Y)$.

$x \backslash y$	0	1	2	3	Row sum
0	0.05	0.05	0.10	0.00	0.20
1	0.05	0.10	0.25	0.10	0.50
2	0.00	0.15	0.10	0.05	0.30
Col. sum	0.10	0.30	0.45	0.15	1.00

Solution. From the margins,

$$\mu_1 = 1.1, \qquad \sigma_1^2 = 1.7 - (1.1)^2 = 0.49, \qquad \sigma_1 = 0.7,$$

$$\mu_2 = 1.65, \qquad \sigma_2^2 = 3.45 - (1.65)^2 = 0.7275, \qquad \sigma_2 = \sqrt{0.7275} = 0.8529.$$

Summing $xy\,p(x,y)$ over the cells with $x, y, p \neq 0$,

$$E(XY) = 0.1 + 0.5 + 0.3 + 0.3 + 0.4 + 0.3 = 1.9.$$

Hence $\mathrm{Cov}(X,Y) = 1.90 - 1.65 \times 1.1 = 0.085$ and

$$\mathrm{Corr}(X,Y) = \frac{0.085}{0.70 \times 0.8529} = 0.1424.$$

Theorem 3-11 (Properties of covariance and correlation)

For constants a, b, c, d:
(1) $\mathrm{Cov}(aX+b,\, cY+d) = ac\ \mathrm{Cov}(X,Y)$;
(2) $\mathrm{Corr}(aX+b,\, cY+d) = \mathrm{Corr}(X,Y)$ if $ac > 0$, and $-\mathrm{Corr}(X,Y)$ if $ac < 0$.

Proof. [Supplement] (1) $E(aX+b) = a\mu_1 + b$ and $E(cY+d) = c\mu_2 + d$, so the centred variables are $a(X-\mu_1)$ and $c(Y-\mu_2)$, and

$$\mathrm{Cov}(aX+b, cY+d) = E[ac(X-\mu_1)(Y-\mu_2)] = ac\ \mathrm{Cov}(X,Y).$$

(2) By Theorem 3-8, $\mathrm{sd}(aX+b) = |a|\,\mathrm{sd}(X)$ and $\mathrm{sd}(cY+d) = |c|\,\mathrm{sd}(Y)$. With (1),

$$\mathrm{Corr}(aX+b, cY+d) = \frac{ac\ \mathrm{Cov}(X,Y)}{|a||c|\,\mathrm{sd}(X)\,\mathrm{sd}(Y)} = \frac{ac}{|ac|}\,\mathrm{Corr}(X,Y),$$

and $ac/|ac| = \pm 1$ according to the sign of ac. ■

Theorem 3-12 (Variance of a sum of two random variables)

(1) $\mathrm{Var}(X+Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) + 2\,\mathrm{Cov}(X,Y)$.
(2) $\mathrm{Var}(X-Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X,Y)$.

Proof. [Supplement] $E(X \pm Y) = \mu_1 \pm \mu_2$ by Theorem 3-9(2), so $(X \pm Y) - E(X \pm Y) = (X-\mu_1) \pm (Y-\mu_2)$. Squaring and taking expectations (Theorem 3-9(2)),

$$\mathrm{Var}(X \pm Y) = E[(X-\mu_1)^2] + E[(Y-\mu_2)^2] \pm 2E[(X-\mu_1)(Y-\mu_2)],$$

which is $\mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$. ■

Theorem 3-13 (Independent random variables)

If X and Y are independent, then
(1) $E(XY) = E(X)E(Y)$;
(2) $\mathrm{Cov}(X,Y) = 0$ and $\mathrm{Corr}(X,Y) = 0$;
(3) $\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$.

Proof. (1) By Theorem 3-9(1) and $p(x,y) = p_1(x)p_2(y)$,

$$E(XY) = \sum_x \sum_y xy\,p(x,y) = \sum_x \sum_y xy\,p_1(x)p_2(y) = \sum_x x\,p_1(x) \sum_y y\,p_2(y) = E(X)E(Y).$$

(2) By Theorem 3-10, $\mathrm{Cov}(X,Y) = E(XY) - E(X)E(Y) = 0$, hence also $\mathrm{Corr}(X,Y) = 0$.

(3) Insert $\mathrm{Cov}(X,Y) = 0$ into Theorem 3-12. ■